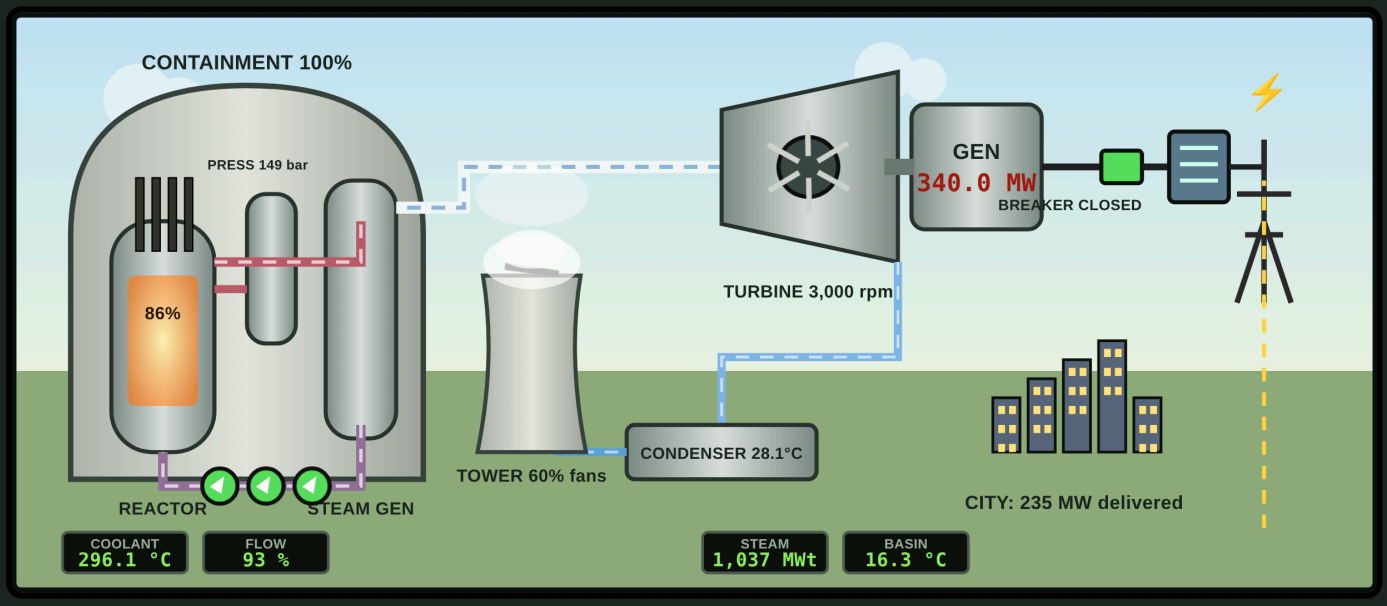
OPERATOR'S FIELD GUIDE

NUCLEAR POWER GRID SIMULATOR

How to run your plant, keep the lights on, stay out of trouble and make money. Short pages, big pictures, no boring walls of text.



Pages	Time to read	Reminder
Quick start, buttons, SCRAM, routines, fixes	About 10 minutes	Everything is FICTIONAL. Real plants are far more complex.

CHAPTER 1

YOUR JOB IN ONE MINUTE

You run a power company. Your reactor makes heat, the heat makes steam, steam spins a turbine, the turbine makes electricity, and customers pay for it. Keep it **safe**, keep it **running**, keep it **profitable**.

Reactor (heat) → **Coolant loop** → **Steam generator** → **Turbine** → **Generator (MW)** → **Grid & customers** → **$$$**

Leftover heat goes to the **condenser** and then the **cooling tower**. If you cannot get rid of heat, power drops and things get hot.

YOUR FIRST 5 MINUTES

1. **Press 1x (or 2x).** The game starts PAUSED. Nothing moves until you pick a speed. You can pause any time to think.
2. **Open Overview.** Check cash, net MW, reactor %, alarms. Green numbers are good, amber means watch it, red means act.
3. **Set the power you want.** On Turbine Hall or Control Room, move **Power Set Point**. With rods on AUTO the reactor follows by itself.
4. **Open Electricity Market & Customers.** Make sure you actually have buyers (contracts) for the power you make.
5. **Check Finance daily.** Revenue, expenses, receivables. Profit keeps you alive.

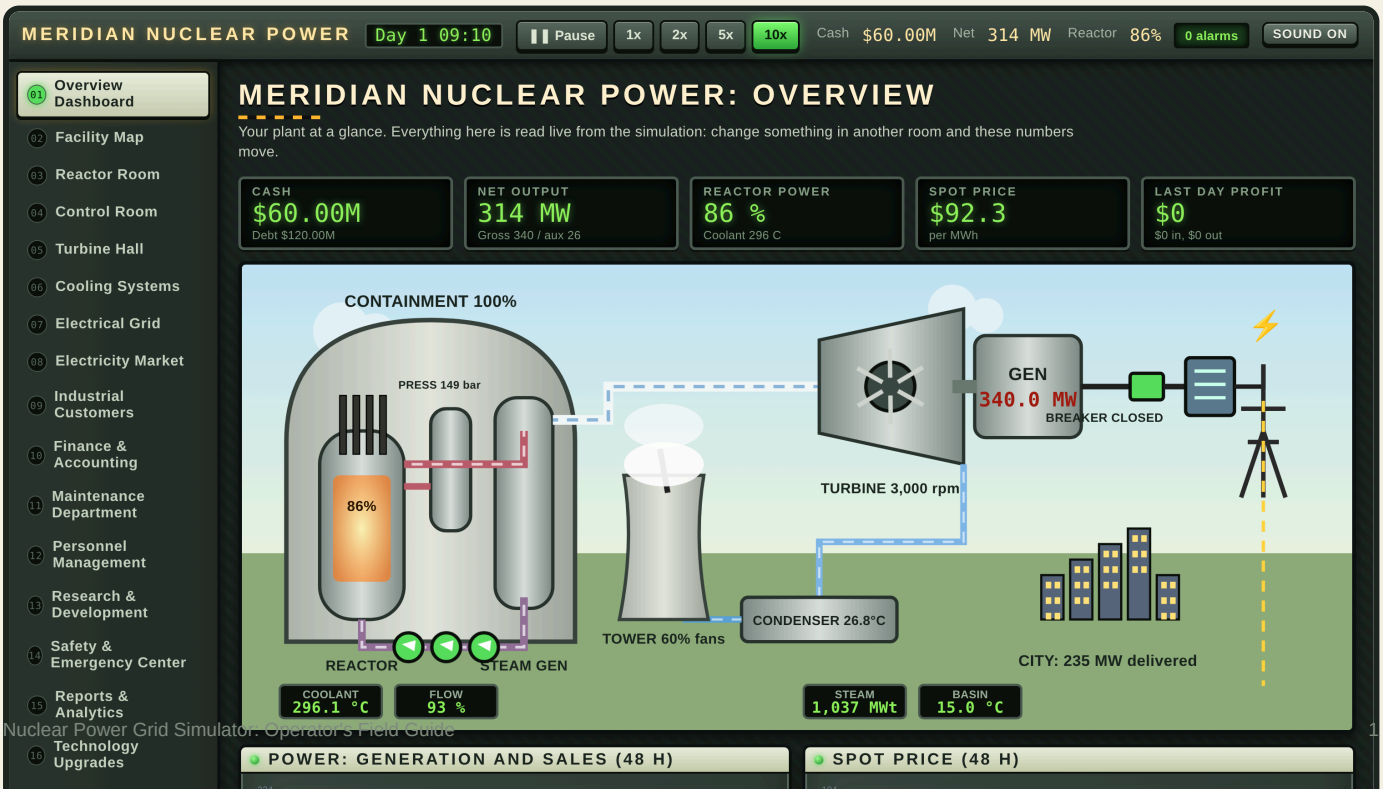
> **PRO TIP** Use 1x or 2x while learning. Reactors react slowly: when you move a control, wait a few minutes of game time before moving it again.

THE 3 SCORES THAT MATTER

S — Safety
No SCRAMs, no ignored alarms, equipment in good condition.

$ — Money
Sell more than you spend. Cash below about $3M turns red.

R — Reputation
Deliver what you promised. Blackouts and failures hurt it.



CHAPTER 2

WHAT EVERY CONTROL DOES



CONTROL	WHAT IT DOES	USE IT WHEN
Rod Demand slider	Control rods absorb neutrons. More withdrawn = more reactor power. Moves slowly, like the real thing.	Only in **MAN** mode. Small steps, then wait.
Rod Control MAN / AUTO	AUTO: rods follow your turbine power set point. MAN: you drive the reactor yourself.	Keep on **AUTO** unless you are practising or something misbehaves.
Power Set Point	How many MW you ask the turbine to make (max about 420 with upgrades, less if worn).	Match your customers' demand, with a little spare.
Tower Fans	More fans = colder condenser = better efficiency, but fans use power (cost grows fast with speed).	Hot weather: raise. Cold, quiet day: lower to save money.
Pump buttons (green/red power)	Start or stop each coolant pump. Pumps move heat out of the reactor. Pump speed sets how much flow.	Green = running. Never run a hot reactor with all pumps off.
Protection NORMAL / BYPASS	NORMAL: the plant trips itself when limits are crossed. BYPASS turns that safety net off.	Leave on NORMAL. Bypass is locked on Beginner and Normal.
Grid Breaker CLOSED / OPEN	Connects your generator to the grid. Open = you sell nothing.	CLOSED normally. Open only in a grid emergency.
Trip / Reset Turbine	Trip stops the turbine fast. Reset lets it start again once it is safe.	High vibration or a turbine problem.
Reset SCRAM	Re-arms the reactor after a SCRAM, when conditions allow.	After the cause is found and fixed.
Acknowledge (ack)	Silences an alarm. It does NOT fix the problem.	After you understand the alarm.

SPEED BUTTONS (TOP BAR)

Pause freezes time. **1x / 2x / 5x / 10x** run the game faster. If something goes wrong, drop to 1x or Pause. Fast speed is for quiet, stable days.

PRO TIP Colours: green running / OK, amber careful, red danger or off. Dials turn their numbers amber then red as limits get close.

CHAPTER 3

SCRAM: THE BIG RED BUTTON



SCRAM = emergency shutdown. All control rods drop in at once, the chain reaction stops fast, and reactor power falls toward zero. It is the "stop everything, safely" button.

PRESS IT WHEN

- Coolant temperature is climbing toward **335 C** and you cannot stop it
- Pressure is heading to **180 bar**
- All pumps are lost or flow is near zero with the reactor running
- Reactor power is racing above **125%**
- You are unsure and things are getting worse fast

DO NOT PRESS IT FOR

- A single amber warning you have not looked at yet
- Normal power changes (use the set point instead)
- Low cash or a bad market day
- "Just to see" on a profitable day (it costs money)

WHAT HAPPENS AFTER

1. Rods drop → **2. Power falls** → **3. Decay heat remains** → **4. Keep pumps running** → **5. Find cause, fix, reset**

The reactor still makes heat

Even when shut down, leftover "decay heat" needs cooling. **Keep the pumps ON** after a SCRAM.

You lose income

No power means unmet contracts, penalties and reputation loss. Restarting takes time. So SCRAM is for real danger, not for convenience.

Restart is slow

A poison called **xenon** builds up after shutdown and can make restarts hard for a while. Be patient, raise rods gently.

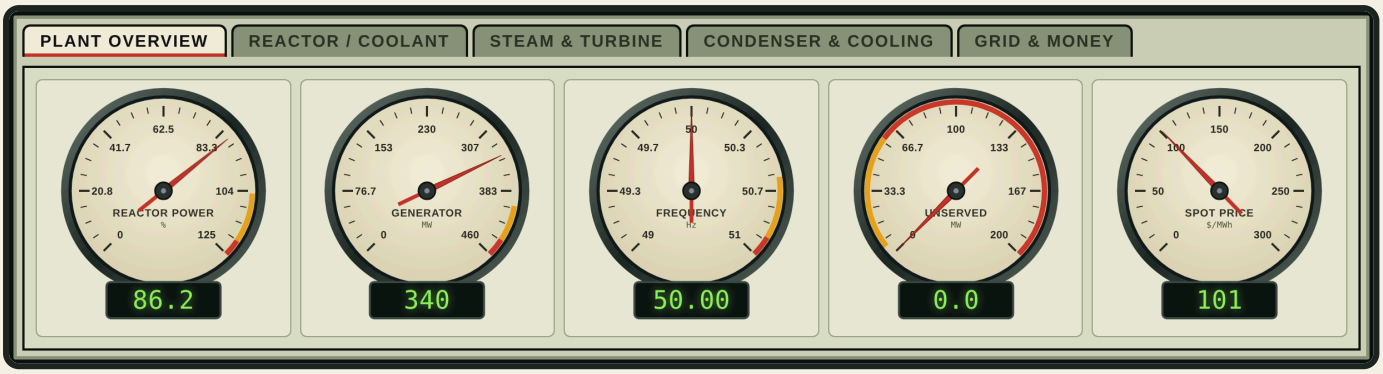
Automatic SCRAMs

With Protection NORMAL the plant may SCRAM itself when limits are crossed. Read the SCRAM reason in the alarm list and fix that first.

WATCH OUT Real nuclear plants have far stricter procedures. This game's limits are simplified and fictional: treat them as practice, not as real-world guidance.

CHAPTER 4

READING THE DIALS



Cream dials show live numbers. The coloured arc at the edge is the danger zone: **amber** = getting close, **red** = trip territory. The green digits below each dial give the exact value. Use the **tabs** to flip between plant overview, reactor, steam and turbine, cooling, and grid and money.

LIMITS TO REMEMBER

DIAL	HEALTHY	WARNING	AUTO-TRIP (GAME LIMIT)
Coolant temperature	about 290 to 310 C	above 322 C	335 C
Pressure	about 140 to 160 bar	above 170 bar	180 bar
Reactor power	0 to 100 %	above 105 %	125 %
Coolant flow	near 100 % at power	below 60 % while above 30 % power	15 %
Turbine vibration	under 6 mm/s	above 6	around 8.5, turbine trips
Condenser temperature	under 45 C	above 45 C (efficiency drops)	n/a
Grid frequency	near 50.0 Hz	below 49.7 Hz	big imbalance = trouble

ANNUNCIATOR (THE WARNING TILES)

Tiles light amber for warnings and flash red for critical items such as SCRAM, low flow, turbine trip and protection bypass. If a tile lights, find the matching dial.

PRO TIP Watch the **trend**, not just the number. A temperature rising slowly by itself is still a problem: act early with small moves.

CHAPTER 5

WHAT TO DO AND WHEN

1

Every few minutes (game time)

- Glance at the alarm chip (top right): green = quiet
- Reactor %, coolant temp, pressure
- Net MW vs customer demand

2

Every hour

- Spot price on the Market page
- Cooling: condenser and basin temperatures
- Weather: heat waves need more fans

3

Every day

- Finance: profit, receivables, cash
- Maintenance: equipment condition and risk
- Staff and shifts, fuel, waste
- News and customer messages

4

Every week or month

- Renew or negotiate contracts
- Plan upgrades and research
- Planned maintenance or outage
- Repay or take loans carefully

OPERATING THE PLANT

A **Match output to demand.** Too little power: customers are unserved and you pay penalties. Too much with no buyers: you waste fuel or sell cheap on the spot market.

B **Change power in small steps.** Move the set point 20 MW, wait, check temperatures, then continue.

C **Take care of the cooling.** Hot day? Raise fans before the condenser passes 45 C. Every degree hotter costs efficiency and money.

D **Maintain before it breaks.** Equipment wears out. Repair when condition drops, schedule work in low-demand hours, keep spare parts in the warehouse.

E **Keep enough staff.** Operators, engineers and technicians have minimums. Too few staff means slower fixes and more mistakes.

PRO TIP Best habit: Pause, look at the alarm list, fix the biggest problem first, then resume. It is not cheating, it is how good operators work.

CHAPTER 6

IF THIS, THEN DO THAT

PROBLEM	LIKELY CAUSE	WHAT TO DO
HI TEMP coolant rising	Too much power, low flow, hot condenser	Lower set point, raise pump speed, raise tower fans. Still rising? SCRAM.
HI PRESS	Coolant too hot, heat not being removed	Reduce power, check pumps and turbine. Near 180 bar: SCRAM.
LO FLOW	Pump stopped or failed	Start another pump, raise speed, cut power. All lost: SCRAM and keep trying to restore flow.
VIBRATION	Worn turbine or bearings	Reduce load, order maintenance. Above the limit the turbine trips.
TURB TRIP	Vibration, grid loss, or you tripped it	Let rods run back, find the cause, fix it, then Reset Turbine.
HI COND	Hot weather, tower worn or fans low	Raise fans. Repair the tower. Lower power if needed.
UNSERVED	Demand above what you deliver, or a line fault	Raise the set point, check the supply lines on the World Map, buy backup power.
LO FREQ / grid trouble	Grid short of power	Make more power if you can. Watch for breaker trips.
LOW CASH	Costs above income	Cut fan and maintenance spending carefully, chase receivables, sell more power. Avoid taking loans you cannot repay.
PROT BYPASS	Protection turned off	Switch Protection back to NORMAL right away.

DO AND DON'T

DO

- Pause to think
- Make small, patient moves
- Fix causes, not just alarms
- Save the game before risky experiments
- Read the in-game "why?" tips

DON'T

- Spam the ack button
- Run at 10x when something is red
- Bypass protection to "win"
- Promise more power than you can make
- Ignore maintenance until it breaks

CHAPTER 7

MONEY, CUSTOMERS AND THE LIVING WORLD

Contracts vs the spot market

Contracts give steady income at an agreed price but you must deliver. The **spot market** price changes by the hour: sell spare power there when prices are high.

Costs that add up

Fuel, water, wages, maintenance, transmission and waste. Costs come from real activity, so a stopped plant still pays wages.

Accrual accounting

Revenue counts when you deliver, but customers pay later (receivables). Cash and profit are different numbers. Watch cash.

Loans

Borrowing helps you grow, but interest is charged daily and repayment comes monthly. Borrow only for things that earn money.

THE WORLD MAP



- Your plant sells to customers across **five regions**. Lines show power flowing, and thicker or brighter means more power.
- **Supply lines have limits and can fail** in storms or when worn. A broken line can leave a customer unserved.
- **Competitors** add power to the market, so the spot price drops when they are strong and rises when they have outages.
- **The economy** moves through boom, slowdown and recession and changes demand, interest and fuel prices.
- Use **Procurement** for parts and fuel, **Construction** for upgrades (needs permits and money), and **Batteries** to store cheap power for peak prices.
- The **Regulator** checks waste handling and reports. Keep records clean.

CHAPTER 8

CHEAT SHEET

START OF GAME

- Press 1x
- Rods on AUTO, Protection NORMAL
- All pumps green
- Breaker CLOSED
- Set point to match demand
- Fans about 50 to 60 %

GOLDEN RULES

- Slow and steady beats fast and sorry
- Cooling is power
- Keep pumps running, even after SCRAM
- Alarms first, money second
- Pause is free

SCRAM IF

- Coolant heading to 335 C
- Pressure heading to 180 bar
- No flow with the reactor running
- Power above 125 % and rising

MAKE MORE MONEY

- Sign contracts you can really deliver
- Sell spare MW on the spot market at high prices
- Keep efficiency up with cold condenser
- Fix things early to avoid big failures

GLOSSARY

MW	Megawatt: how much electric power you make.
MWt	Megawatt thermal: heat power. About 3 MWt of heat makes 1 MW of electricity.
Control rods	Rods that soak up neutrons to slow the reaction.
Coolant	Water that carries heat out of the reactor.
Steam generator	Passes heat from coolant to a separate steam loop.
Condenser	Turns used steam back into water; the colder, the more efficient.
Decay heat	Heat a reactor keeps making after shutdown.
Xenon	A reaction byproduct that makes restarts harder for a while.
Spot price	The changing price for power sold on the open market.
Receivables	Money customers owe you but have not paid yet.

WATCH OUT This is a fictional learning game. Numbers, limits and procedures are simplified and are not real-world safety guidance.

GOOD LUCK, OPERATOR.